

# Supplementary Material: Evidence-Gated Cloud–Edge LLM Serving with Delayed Activation

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**Abstract:** This supplementary document provides the technical derivations, protocol details, secondary decision analyses, calibration details, and retrospective validation material supporting the main manuscript.

**Keywords:** cloud–edge computing; LLM serving; supplementary material; simulation optimization; risk certification

## 1 Scope, notation, and evidence hierarchy

The main manuscript addresses a finite policy-selection problem for delayed cloud activation in privacy-constrained cloud–edge LLM serving. The policy space is

$$\mathcal{A}_q = \{(\alpha_i, \alpha_j) : 0 \leq j \leq i \leq q - 1\}, \quad |\mathcal{A}_q| = \frac{q(q+1)}{2},$$

and the principal experiments use  $q = 19$ , giving 190 activation–release policies. The local state  $X(t) \in [0, K]$  is a modeled workload-pressure index,  $N(t) \in \{0, 1, 2\}$  is the remote-service phase, and  $(\alpha_{\text{on}}, \alpha_{\text{off}})$  controls activation and release.

The evidence is intentionally layered. Search-stage simulation nominates a policy; independent holdout simulation measures selection quality; later-pool replays test an operational statement; exhaustive evaluation diagnoses whether abstention reflects search error or the tested operating envelope. These objects are not pooled.

Table 1: Evidence layers and the claims they support.

| Layer                       | Selection evidence                    | Evaluation evidence               | Interpretation                                                    |
|-----------------------------|---------------------------------------|-----------------------------------|-------------------------------------------------------------------|
| Synthetic development       | High-fidelity training streams        | Independent holdout streams       | Relative policy-selection quality under a fixed budget            |
| Locked 59-scenario study    | Prespecified synthetic scenarios      | Independent holdout streams       | Scenario-level reliability under the declared excess threshold    |
| Original trace study        | Earlier chronological BurstGPT pool   | Later chronological replay pool   | Simulator-conditional post-selection risk evidence                |
| Operating-envelope campaign | Complete 190-policy surfaces          | Enlarged replay batches           | Search error, sample insufficiency, or modeled infeasibility      |
| Fresh low-load confirmation | Configuration fixed after exploration | Two disjoint 1,200-replay batches | Reproducibility of positive decisions at the fixed interior point |
| Seven-day replication       | First five trace days                 | Following two trace days          | Robustness of the qualitative high-load/low-load contrast         |
| Hardware calibration        | Direct service telemetry              | Leave-one-level-out prediction    | Reduced-order service-law shape and parameter range               |

Throughout the supplement, “certification” means that the exact replay-level risk rule is satisfied under the declared simulator and replay generator. It does not mean production-system safety, safe exploration, or global optimality. “Abstention” means that the prespecified evidence rule is not satisfied.

## 2 Stationary diffusion screening model

### 2.1 Moment approximation and reflected process

The compound-jump simulator is the reference fidelity. For cloud phase  $n \in \{0, 1, 2\}$ , the set of request classes offered locally is

$$\mathcal{K}_0 = \mathcal{K}_1 = \{H, L\}, \quad \mathcal{K}_2 = \{H\}.$$

With class arrival rates  $\nu_k$ , pressure-jump means  $m_k$ , and second moments  $m_k^{(2)}$ , the first two infinitesimal moments are

$$b_n(x) = -s_n(x) + \sum_{k \in \mathcal{K}_n} \nu_k m_k, \quad a_n = \sum_{k \in \mathcal{K}_n} \nu_k m_k^{(2)}.$$

The low-fidelity process is

$$dX(t) = b_{N(t)}(X(t)) dt + \sqrt{a_{N(t)}} dB(t) + dL_0(t) - dL_K(t),$$

where  $L_0$  and  $L_K$  are nondecreasing regulators acting only at the boundaries. Reflection is an analytical and numerical device of the screen. It differs from the reference simulator, where an entire request that would exceed  $K$  is rejected and leaves the state unchanged.

The moment approximation preserves phase-dependent mean input and local variance. It removes overshoot geometry, jump skewness, whole-request rejection, and the eligible-overflow activation trigger. These omissions explain why the diffusion can be useful for ranking while remaining inaccurate for absolute blocking and delay.

### 2.2 Stationary forward equations and flux

For fixed policy  $a = (\alpha_{\text{on}}, \alpha_{\text{off}})$ , let

$$\boldsymbol{\pi}^a(x) = (\pi_0^a(x), \pi_1^a(x), \pi_2^a(x))$$

be the stationary phase-density row vector. Define

$$B(x) = \text{diag}(b_0(x), b_1(x), b_2(x)), \quad A = \text{diag}(a_0, a_1, a_2).$$

On each open interval separated by the two thresholds,

$$-\frac{d}{dx} [\boldsymbol{\pi}^a(x) B(x)] + \frac{1}{2} \frac{d^2}{dx^2} [\boldsymbol{\pi}^a(x) A] + \boldsymbol{\pi}^a(x) T_a(x) = \mathbf{0}.$$

The phase- $n$  stationary flux is

$$\mathcal{J}_n^a(x) = b_n(x) \pi_n^a(x) - \frac{1}{2} \frac{d}{dx} [a_n \pi_n^a(x)].$$

Normal reflection gives

$$\mathcal{J}_n^a(0) = 0, \quad \mathcal{J}_n^a(K) = 0.$$

At each policy threshold, the forward equations are interpreted weakly: probability mass is conserved, the appropriate flux balance is retained, and no artificial point mass is inserted. The density is normalized by

$$\sum_{n=0}^2 \int_0^K \pi_n^a(x) dx = 1.$$

### 2.3 Invariant-distribution assumptions

The stationary characterization uses standard assumptions for reflected switching diffusions.

**Proposition 1** (Fixed-policy invariant distribution). *Fix  $a \in \mathcal{A}_{19}$ . Assume that the reflected switching diffusion is well posed and Feller on the compact state space  $[0, K] \times \{0, 1, 2\}$ ; that the phase diffusion coefficients are strictly positive; that the drifts are bounded and piecewise Lipschitz; that switching intensities are bounded and piecewise continuous; that the phase graph communicates; and that the joint process is irreducible. Then the process has a unique invariant probability distribution. Its interior density satisfies the stationary equations above, zero normal flux at 0 and  $K$ , and weak transmission balance at the thresholds.*

Compactness and the Feller property support existence. Strict ellipticity, phase communication, and irreducibility provide the additional structure needed for uniqueness. The proposition records the basis of the numerical screen and is not claimed as a new theorem.

## 2.4 Performance measures

Let

$$P_n^a = \int_0^K \pi_n^a(x) dx, \quad p^a(x) = \sum_{n=0}^2 \pi_n^a(x).$$

The cloud-active fraction and mean local pressure are

$$R_{C,D}(a) = P_2^a, \quad \bar{X}_D(a) = \int_0^K x p^a(x) dx.$$

Using the jump-tail functions  $\bar{F}_k(y) = \mathbb{P}(J_k > y)$ , the approximate loss probabilities are

$$B_{H,D}(a) = \sum_{n=0}^2 \int_0^K \bar{F}_H(K-x) \pi_n^a(x) dx,$$

and

$$B_{L,D}(a) = \sum_{n=0}^1 \int_0^K \bar{F}_L(K-x) \pi_n^a(x) dx.$$

For the nominal Poisson model, arrival-seen probabilities are interpreted through PASTA. This interpretation is not transferred unchanged to trace-derived replay.

If  $\Lambda_{\text{adm},D}(a)$  is the approximate stationary local-admission rate, the pressure-delay proxy is

$$W_D(a) = \frac{\bar{X}_D(a)}{\Lambda_{\text{adm},D}(a)}.$$

Because  $X(t)$  is an aggregate pressure index rather than a literal request count,  $W_D$  is a calibrated proxy rather than an exact queueing-delay identity. The low-fidelity objective is

$$J_D(a) = c_W W_D(a) + c_H B_{H,D}(a) + c_L B_{L,D}(a) + c_C R_{C,D}(a).$$

## 2.5 Continuous-time Markov-chain discretization

For each policy, the pressure interval is discretized on a uniform grid. A nearest-neighbour continuous-time Markov chain matches the local drift and variance while preserving nonnegative off-diagonal rates. Phase transitions are added through the piecewise generator  $T_a(x)$ . If  $Q_a$  is the finite-state generator, the stationary vector  $p_a$  solves

$$p_a Q_a = 0, \quad p_a \mathbf{1} = 1.$$

The implementation checks generator-row balance, probability normalization, nonnegativity, and the stationary residual.

Table 2: Audited numerical checks for the stationary screen.

| Quantity                                    | Reported value         |
|---------------------------------------------|------------------------|
| Maximum probability-mass error              | $2.22 \times 10^{-16}$ |
| Maximum stationary residual                 | $1.43 \times 10^{-12}$ |
| Negative stationary probabilities           | None                   |
| Median change, 321 to 641 grid points       | 0.49%                  |
| 90th-percentile change                      | 0.97%                  |
| Maximum change                              | 3.07%                  |
| Median absolute metric error vs. jump model | 24.3%                  |
| 90th-percentile absolute metric error       | 64.1%                  |

The first six quantities establish numerical accuracy for the discretized diffusion. The last two measure fidelity to the compound-jump model and justify residual correction. Numerical solution accuracy is not model validation.

### 3 Guarded multi-fidelity search

#### 3.1 Mechanistic trend and residual Gaussian process

After iteration  $t$ , let  $E_t \subset \mathcal{A}_{19}$  be the high-fidelity policies already evaluated. The current affine calibration is

$$(\hat{b}_{0,t}, \hat{b}_{1,t}) \in \arg \min_{b_0, b_1} \sum_{a \in E_t} [\bar{Y}_J(a) - b_0 - b_1 J_D(a)]^2.$$

The mechanistic trend is

$$m_{0,t}(a) = \hat{b}_{0,t} + \hat{b}_{1,t} J_D(a),$$

and the high-fidelity residual is

$$r_t(a) = \bar{Y}_J(a) - m_{0,t}(a).$$

A Matérn-5/2 Gaussian process models  $r_t(a)$  over normalized threshold coordinates. With posterior residual mean  $\hat{r}_t(a)$  and standard deviation  $s_t(a)$ , the next policy minimizes

$$\text{LCB}_t(a) = m_{0,t}(a) + \hat{r}_t(a) - \beta_t s_t(a), \quad \beta_t = 2 + 0.25 \log(1 + t).$$

The diffusion contributes a structured mean surface; it is not a feasibility or safety model.

#### 3.2 Structural anchors

The initial design contains representative policies near the lower corner, upper diagonal, upper off-diagonal corner, and two intermediate regions, together with the three policies having the smallest diffusion objectives. Duplicates are removed. If fewer than eight distinct policies remain, maximin distance in normalized threshold coordinates fills the initial set.

These points are called structural guards because they reduce the chance that a small initial design omits important parts of the triangular policy space. They do not enforce blocking constraints. The same anchors are retained in the guarded-GP ablation, making that ablation the cleanest test of the diffusion residual.

#### 3.3 Fixed budget and nomination rule

Each evaluated policy receives 20 common-random-number training replications. Sequential acquisition stops after 20 distinct policies. The nomination is the evaluated policy with the lowest observed high-fidelity training mean:

$$\hat{a} \in \arg \min_{a \in E_T} \bar{Y}_J(a), \quad |E_T| = 20.$$

The nomination receives 30 independent holdout replications. The budget is

$$N_{\text{DR-BGS}} = 20 \times 20 + 30 = 430,$$

whereas exhaustive training selection with the same holdout allocation uses

$$N_{\text{exhaustive}} = 190 \times 20 + 30 = 3830.$$

The reduction is 88.77%.

#### 3.4 Algorithmic sequence

For each scenario, the implementation follows this fixed sequence:

1. evaluate the diffusion objective on all 190 policies;
2. form the eight-policy structural initial design;
3. run 20 common-random-number high-fidelity replications per initial policy;
4. fit the affine trend and residual GP;
5. select one unevaluated policy by the lower-confidence rule;
6. evaluate it with the same training replication budget;
7. repeat Steps 4–6 until 20 policies have been evaluated;
8. nominate the evaluated policy with the smallest observed training mean; and
9. evaluate the nomination on 30 disjoint holdout replications.

Five fixed initialization seeds measure sensitivity to maximin tie-breaking and sequential search. Kernel family, budget, and acquisition schedule are not retuned per scenario.

### 3.5 Comparators and ablations

The common 20-policy budget is assigned to random-subset ranking and selection, high-fidelity neighbourhood search, diffusion top-20, standard GP Bayesian optimization, guarded GP without a diffusion residual, and a breadth-first two-stage ranking-and-selection procedure. The main comparison uses independent holdout excess relative to exhaustive training selection, not a holdout oracle.

The same-anchor guarded GP reaches 0.878% maximum excess, compared with 0.348% for DR-BGS. This indicates that structural coverage supplies most of the observed reliability at the 20-policy budget; the diffusion residual adds a smaller improvement.

## 4 Independent certification and retrospective audits

### 4.1 Search-stage objective and fallback

For the trace-derived study, the search-stage cost is

$$C(a) = R_C(a) + 0.15W(a),$$

with nominal training-mean limits

$$B_H(a) \leq 0.02, \quad B_L(a) \leq 0.05.$$

If one or more evaluated policies satisfy both limits, the feasible policy with the smallest  $C(a)$  is nominated. Otherwise, the search minimizes

$$C(a) + 50[B_H(a) - 0.02]_+ + 20[B_L(a) - 0.05]_+.$$

The fallback guarantees a nomination for independent evaluation. It does not declare the nomination feasible and does not alter the certification rule.

### 4.2 Replay-level Bernoulli indicators

Certification uses  $n_{\text{cert}} = 150$  later-pool replays that are disjoint from search and holdout. For replay  $r$ ,

$$V_{H,r} = \mathbf{1}\{\widehat{B}_{H,r} > 0.02\}, \quad V_{L,r} = \mathbf{1}\{\widehat{B}_{L,r} > 0.05\}.$$

The violation counts are

$$x_H = \sum_{r=1}^{n_{\text{cert}}} V_{H,r}, \quad x_L = \sum_{r=1}^{n_{\text{cert}}} V_{L,r}.$$

For  $j \in \{H, L\}$ , the one-sided Clopper–Pearson upper bound is

$$U_j = \text{BetaQuantile}(1 - \alpha_j; x_j + 1, n_{\text{cert}} - x_j),$$

with the conventional endpoint definition when  $x_j = n_{\text{cert}}$ . The allocation  $\alpha_H = \alpha_L = 0.025$  gives at least 95% simultaneous coverage by Bonferroni.

The policy is certified only if

$$U_H \leq 0.10, \quad U_L \leq 0.10.$$

The 2% and 5% values are within-replay blocking limits. The 0.10 values are limits on the probabilities that a replay violates those blocking limits.

### 4.3 Diagnosing abstention

Abstention can have three principal explanations:

1. the nominated policy is poor;
2. no policy in the tested family satisfies the rule at the operating point; or
3. the certification batch is too small to support the claim.

The exhaustive 190-policy audit addresses the first two possibilities. The enlarged 1,200-replay batches address the third. At the original loads, no policy among 2,280 policy–scenario pairs certifies, and increasing the batch does not change the decision. Under the declared model, universal abstention is therefore not a consequence of a missed candidate or the original batch size.

Table 3: Interpretation of the certification rule.

| Outcome                             | Supported interpretation                                                                                                                                           |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Both bounds at most 0.10            | Under the declared later-pool replay generator, the evidence supports the two replay-level violation-probability limits with at least 95% simultaneous confidence. |
| At least one bound above 0.10       | The prespecified evidence rule is not satisfied; the procedure abstains.                                                                                           |
| Certification does not imply        | Safe search, global optimality, production safety, correct privacy labels, or validity after drift.                                                                |
| Abstention does not by itself prove | That the policy is unsafe or that the search failed.                                                                                                               |

#### 4.4 Paired noninferiority audit

A separately locked retrospective analysis compares each nomination with the policy selected by exhaustive high-fidelity training evaluation. Both policies are evaluated on paired later-pool seeds. Prespecified noninferiority margins are applied to the paired metric differences.

The reference requires evaluating all 190 policies and is therefore not an economical operational certificate. The analysis is retained only as a decision-quality audit. It certifies DR-BGS as noninferior in 50 of 60 runs and guarded GP in 33 of 60; every certified run recovers the exhaustive-training-selected policy.

### 5 Detailed experimental chronology

#### 5.1 Synthetic development benchmark

The 36-scenario development benchmark contains 12 maximin factorial cases, 12 maximin Latin-hypercube robustness cases, and 12 additional Latin-hypercube cases. Factors vary offered load, burstiness, jump scale, activation delay, privacy mix, service degradation, jump variability, cloud-use price, release delay, and capacity.

Training uses 20 common-random-number replications per policy. Each replication has a 225-unit warm-up and a 900-unit measurement horizon. Holdout evaluation uses 30 disjoint replications with a 300-unit warm-up and a 1,200-unit measurement horizon. Five initialization seeds produce  $36 \times 5 = 180$  selections per stochastic method.

Let

$$a_{\text{ex,tr}} \in \arg \min_{a \in \mathcal{A}_{19}} \bar{Y}_{\text{tr}}(a).$$

For nomination  $\hat{a}$ , the holdout excess is

$$R_{\text{ex}} = \max \left\{ 0, \frac{\bar{Y}_{\text{hold}}(\hat{a}) - \bar{Y}_{\text{hold}}(a_{\text{ex,tr}})}{\bar{Y}_{\text{hold}}(a_{\text{ex,tr}})} \right\}.$$

The exhaustive reference is selected on training data and evaluated on the same independent holdout surface as the budgeted nomination.

#### 5.2 Independent 59-scenario studies

In the strict locked study, a scenario succeeds only if all five initializations satisfy  $R_{\text{ex}} \leq 1\%$ . DR-BGS succeeds in 57 of 59 scenarios, producing a one-sided 95% lower confidence bound of 89.71%. The two failures have maximum excesses of 1.15% and 3.49%; all 295 runs remain within 5%.

After that result, a separate protocol fixes a 5% practical-indifference threshold before generating a follow-up. All 59 scenarios succeed, the one-sided lower bound is 95.05%, and the maximum excess is 0.941%. The follow-up is not used to rewrite the stricter result.

#### 5.3 Original chronological BurstGPT study

The original study uses the first 100 chronological rows of the official `BurstGPT_1.csv` release. After removing nonpositive token counts, 95 records remain. The first 60 form the search pool and the later 35 form the certification pool.

A moving-block bootstrap of length eight resamples timestamp gaps and token counts. Each replay contains 560 requests. If  $z$  is token count, the pressure jump is

$$J = \min \left\{ 0.25K, \max \left[ 0.10, \frac{3z}{\tilde{z}_{\text{search}}} \right] \right\}, \quad \tilde{z}_{\text{search}} = 912.$$

The public trace does not contain privacy labels. Protected and eligible classes are generated through independent Bernoulli draws with

$$q_H \in \{0.20, 0.50, 0.80\}.$$

The 12 scenarios cross

$$\rho \in \{0.70, 0.90\}, \quad q_H \in \{0.20, 0.50, 0.80\}, \quad \tau \in \{0.5, 2.0\}.$$

The later pool is not used to tune the search, penalties, or certification thresholds.

## 5.4 Operating-envelope exploration and fresh confirmation

After universal high-load abstention, an exploratory campaign evaluates all 190 policies while varying offered load, capacity, mean activation delay, and certification batch size. All failures are retained.

The study finds:

- 0 of 2,280 policy–scenario pairs certify at  $\rho \in \{0.70, 0.90\}$  and  $K = 40$ ;
- reducing mean activation delay to 0.10 certifies 0 of 12 scenarios;
- increasing capacity through  $K = 240$  certifies 2 of 12 scenarios; and
- at baseline capacity, no load-only case certifies at  $\rho \geq 0.10$ .

After the exploration, the interior point  $K = 40, \rho = 0.05$  is fixed before fresh result generation. Six DR-BGS, 20 guarded-GP, and six exhaustive-training pairs pass both disjoint 1,200-replay batches. After cross-method deduplication, all 31 distinct pairs remain confirmed.

## 5.5 Seven-day replication

The locked seven-day protocol uses 11,810 positive-token records from the first five days for search and 6,086 records from the following two days for later certification. The policy space, pressure transformation, risk rule, and method budgets remain unchanged.

At the original loads, none of the 2,280 exhaustive policy–scenario pairs certifies. At the fixed low-load point, 24 method-specific pairs pass both 1,200-replay batches; deduplication leaves 18 distinct confirmed pairs. The largest upper risk bound among these low-load pairs is 0.00307.

## 5.6 Protocol integrity

The archive records scenario definitions, policy grids, random seeds, trace transformations, protocol locks, result files, and content hashes. The seven-day inputs and outputs are covered by immutable provenance checks over tracked lock metadata, locked-input hashes, required artifacts, and the repository checksum manifest. Filesystem modification times are not used as scientific chronology evidence. Experiments without defensible historical pre-result locks remain labeled descriptive or retrospective.

# 6 Secondary decision analyses

## 6.1 Policy-space scaling

The exploratory scaling study uses

$$q \in \{19, 31, 51\},$$

corresponding to 190, 496, and 1,326 policies. Three representative scenarios are exhaustively evaluated at each resolution using six training and ten holdout replications per policy. DR-BGS keeps its fixed 20-policy budget.

Table 4: Exploratory policy-space scaling.

| Policies | Evaluated fraction | Work saved | Median excess | Maximum excess |
|----------|--------------------|------------|---------------|----------------|
| 190      | 10.53%             | 88.70%     | 1.45%         | 5.96%          |
| 496      | 4.03%              | 95.65%     | 0.10%         | 3.15%          |
| 1,326    | 1.51%              | 98.37%     | 0.00%         | 0.00%          |

The result demonstrates that the evaluated fraction decreases rapidly as the triangular grid grows. It does not show that a constant 20-policy budget will remain adequate for every larger problem or scenario family.

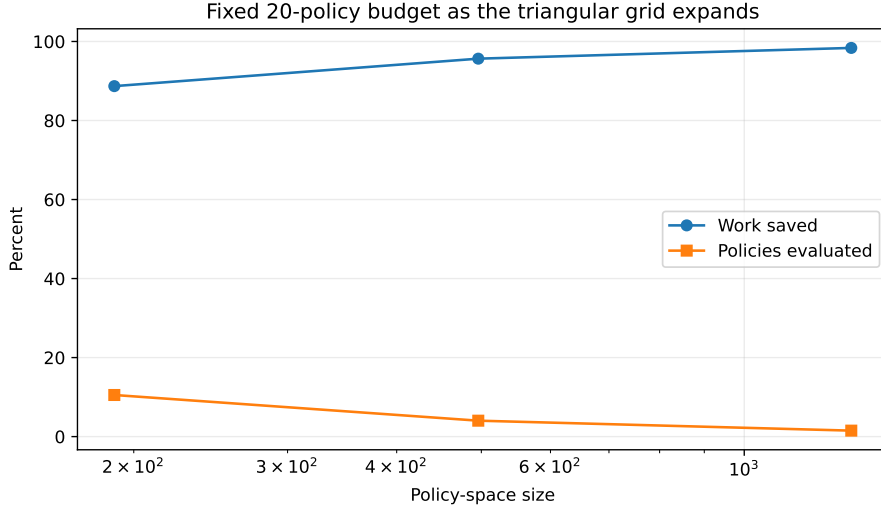


Figure 1: Work reduction and evaluated fraction under a fixed 20-policy high-fidelity budget. The experiment is a computational scaling check rather than a reliability study.

## 6.2 Practical reference policies

Using one prespecified DR-BGS initialization, the selected policies improve the median holdout objective by 30.96% relative to local-only execution, 13.28% relative to  $(0.5K, 0.5K)$ , and 11.79% relative to the fixed  $(0.5K, 0.4K)$  deadband. DR-BGS is better in all 36 scenarios for these three comparisons.

Against the strongest diagonal policy selected on training data, the median difference is  $-0.38\%$ , and DR-BGS is better in 41.67% of scenarios. Strict hysteresis is therefore not uniformly superior to the strongest single-threshold alternative.

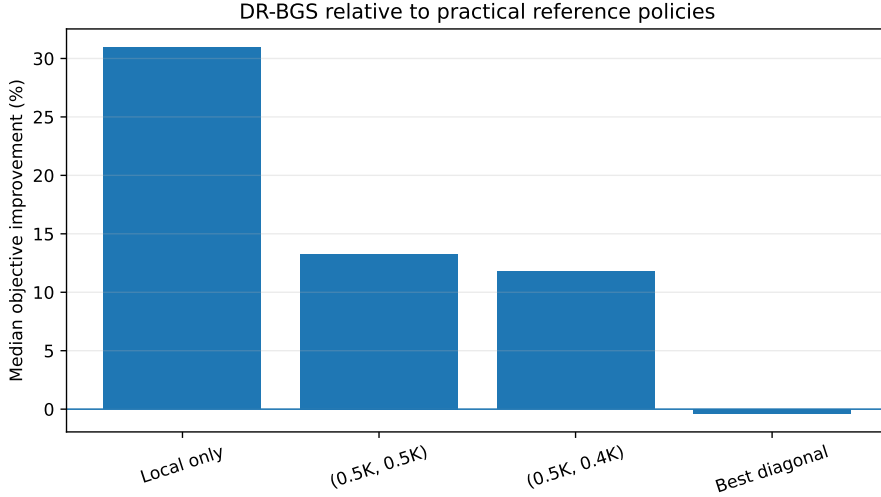


Figure 2: Median objective improvement of DR-BGS relative to practical reference policies. Negative improvement against the strongest diagonal policy indicates no universal strict-hysteresis advantage.

## 6.3 Activation-delay sensitivity

A delay-aware diffusion design evaluated under the actual-delay jump model improves over a policy selected under a near-instantaneous activation assumption by 3.04% in the median, 12.82% at the 90th percentile, and 17.69% at maximum.

The median effect is moderate, but the upper tail shows that ignoring startup delay can materially alter policy quality in some regimes.



Table 5: Benefit of delay-aware policy design under the actual-delay model.

| Summary statistic | Objective improvement |
|-------------------|-----------------------|
| Median            | 3.04%                 |
| 90th percentile   | 12.82%                |
| Maximum           | 17.69%                |

## 6.4 Incremental value of strict hysteresis

Using complete holdout surfaces as a retrospective diagnostic, the best strict-hysteresis policy improves over the best diagonal policy in 41.67% of the original 36 scenarios. The median gain is zero, the 90th percentile is 1.03%, and the maximum is 2.85%.

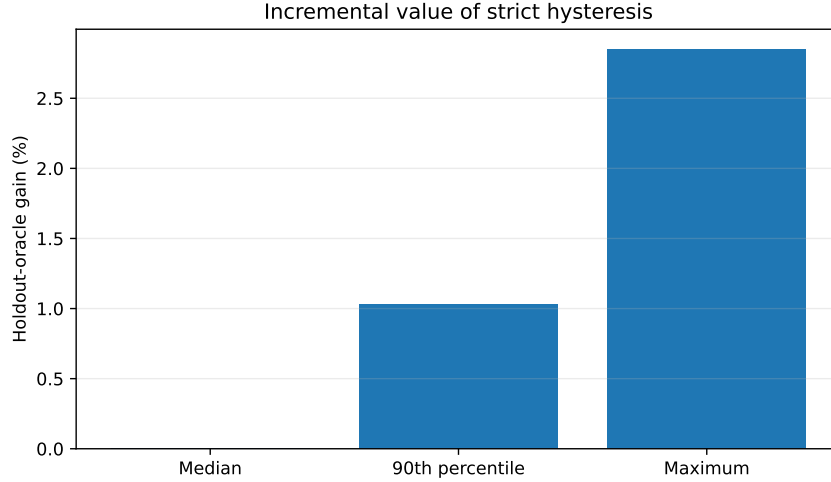


Figure 3: Summary of incremental holdout-oracle gain from admitting a separate release threshold.

The expanded policy space remains useful because it includes both diagonal and strict-hysteresis alternatives and permits data-driven selection. The result does not support a universal superiority claim.

## 6.5 Pareto and minimax-regret policies

The holdout metrics ( $W, B_H, B_L, R_C$ ) are normalized within scenario. A simplex grid of 286 weight vectors with step 0.1, augmented by equal weights, produces 287 stakeholder profiles. For policy  $a$  and profile  $w$ , let  $C_w(a)$  be the normalized weighted cost and  $C_w^*$  its optimum. The minimax-regret policy is

$$a_{\text{MR}} \in \arg \min_a \max_w [C_w(a) - C_w^*].$$

Across 95 scenarios, the median Pareto set contains 118 of 190 policies, and the median number of distinct profile-optimal policies is 13. The minimax policy differs from the equal-weight policy in every scenario and is Pareto efficient on the point-estimate surfaces. Its median maximum normalized regret is 0.462, compared with 1.000 for the equal-weight policy, a reduction of 52.63%. The compromise pays a median normalized penalty of 0.174 under equal weights.

In the median scenario, the minimax policy is more than 5% above the profile optimum for 83.62% of profiles, compared with 19.51% for the equal-weight policy. These results use point estimates and do not include uncertainty in Pareto dominance.



Figure 4: Median maximum normalized regret over 287 stakeholder profiles.

## 7 Service-law calibration detail

### 7.1 Measurement role

The local-service law is

$$s_n(x) = s_{0,n} \exp(-\beta_n x / K).$$

The calibration tests whether an exponential decline is a plausible reduced-order description over an observed hardware range. It does not validate the complete threshold controller.

The principal measurements use Qwen2.5-7B served with vLLM on an NVIDIA RTX 3090 at six concurrency levels. The predictor is mean vLLM GPU KV-cache usage, and the response is effective per-stream decode rate. Hardware, serving configuration, model, request generation, and metric definitions are held fixed across the six levels.

### 7.2 Fit and cross-validation

The fitted law is

$$\hat{\mu}(z) = 53.6138 \exp(-0.034451z).$$

It gives

$$R^2 = 0.9961, \quad \text{RMSE} = 0.789 \text{ tokens/s}.$$

Because the in-sample  $R^2$  is based on the same six operating levels used for fitting, leave-one-level-out refitting is also reported. Its prediction RMSE is 1.299 tokens/s.

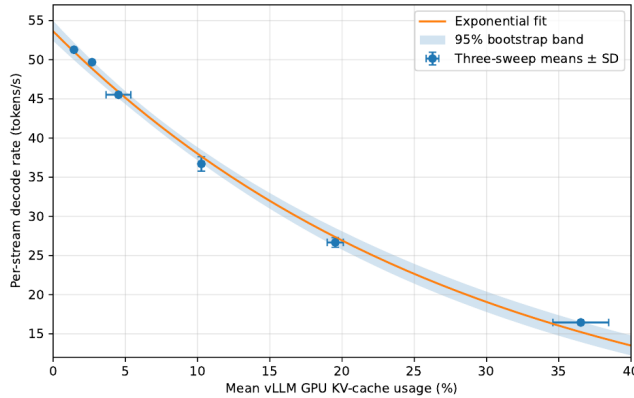


Figure 5: Measured effective decode rate and fitted exponential law for Qwen2.5-7B on the RTX 3090.

### 7.3 Bootstrap and simulator mapping

A 20,000-draw bootstrap gives a normalized observed-range degradation interval of

$$1.159 \leq \beta \leq 1.358.$$

The synthetic benchmark uses

$$0.600 \leq \beta \leq 1.376,$$

which deliberately includes slower degradation and additional structural uncertainty. The fitted coefficient 0.034451 is not compared directly with dimensionless  $\beta$  before mapping the measured  $z$ -range to the normalized state  $x/K$ .

RTX 4090 and Mistral measurements are retained as qualitative checks. They show the same directional decline but are not pooled because their hardware, model, or direct-telemetry protocols differ.

The calibration supports only three limited statements:

1. effective service declines as measured cache-related pressure rises;
2. an exponential law fits the principal RTX 3090 measurements over the observed range; and
3. the simulation range contains the telemetry-derived estimates and additional structural uncertainty.

It does not establish coefficient transfer across GPUs or models, exact physical meaning of  $X(t)$ , or end-to-end controller validity.

## 8 Retrospective analytical and model-form validation

This section documents a post hoc campaign performed after the primary study results were known. It is not a preregistered or prospective replication.

### 8.1 Independent optimizer implementation

The public structural-anchor and Gaussian-process rules were reimplemented from the archived algorithm description and applied directly to the 36 exhaustive policy surfaces.

Table 6: Independent optimizer reanalysis over 180 scenario-seed runs.

| Method      | Within 1% | Within 5% | Maximum excess | Exact recovery |
|-------------|-----------|-----------|----------------|----------------|
| DR-BGS      | 100.0%    | 100.0%    | 0.348%         | 88.9%          |
| Guarded GP  | 100.0%    | 100.0%    | 0.878%         | 86.1%          |
| Standard GP | 95.6%     | 98.9%     | 7.220%         | 78.3%          |

The reimplementations reproduce the reported DR-BGS and guarded-GP worst cases.

### 8.2 Policy-surface ranking and absolute error

Across 36 scenarios, the median diffusion/holdout Spearman correlation is 0.989, with scenario-bootstrap 95% interval 0.977–0.992. Median Kendall correlation is 0.921, and median top-20 overlap is 85.0%. In contrast, median scenario-level absolute objective error is 33.4%.

The diffusion-best policy exactly matches exhaustive training selection in 50.0% of scenarios. Its 90th-percentile and maximum holdout excesses are 5.68% and 13.86%, respectively. The diffusion is therefore suitable as a trend but not as the sole policy selector.

### 8.3 Jump-remainder and boundary indicators

For Gamma jumps with mean  $m$  and coefficient of variation  $c$ ,

$$\mathbb{E}[J^2] = m^2(1 + c^2), \quad \mathbb{E}[J^3] = m^3(1 + c^2)(1 + 2c^2).$$

The interior Taylor remainder satisfies

$$|R_3 f| \leq \frac{M_3}{6} \|f'''\|_\infty,$$

and the capacity-scaled ratio of third- to second-order coefficients is  $M_3/(3KM_2)$ . Its median is 0.075 and its maximum is 0.216. The median protected-jump overflow probability within the final 5% of capacity is 0.484.

Jump coefficient of variation and the third-order indicator have Spearman associations 0.438 and 0.348 with rank degradation. These are diagnostic associations rather than causal estimates.

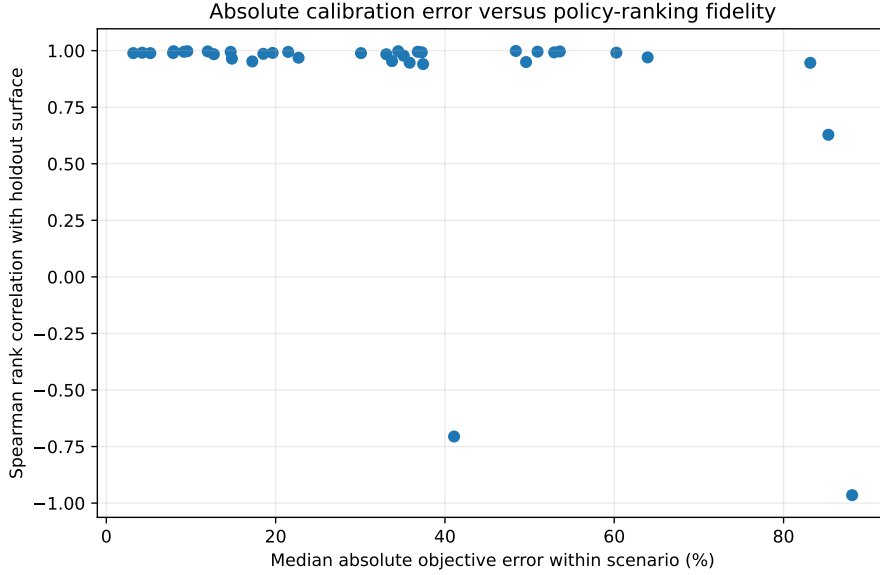


Figure 6: Policy-ranking fidelity versus absolute objective error.

## 8.4 Direct service-law model comparison

The direct RTX 3090 fit is

$$\hat{\mu}(z) = 53.6138 \exp(-0.034451z).$$

Its RMSE is 0.789 tokens/s,  $R^2 = 0.9961$ , AICc 5.16, and leave-one-level-out RMSE 1.299 tokens/s. The 10,000-draw bootstrap intervals are 53.376–53.857 for the intercept and 0.03326–0.03567 for the decay coefficient.

## 8.5 Cross-hardware and cross-model transport

All primary curves are strictly decreasing. Pairwise normalized-curve RMSE ranges from 0.067 to 0.133. Leave-one-domain-out RMSE ranges from 0.103 to 0.131. The shared exponential  $R^2$  is 0.812. A common decline shape is supported, but a common coefficient is not.

## 8.6 Low-fidelity model-form stress

Every tested transformation retains a 100% within-5% rate. Positive affine transformation is invariant. The strongest smooth perturbation has median Kendall correlation 0.944 and maximum excess 4.29%. These are heuristic sensitivity tests rather than physical service-model substitutions.

## 8.7 Independent solver triangulation

Eighteen unique policy cases from six scenario types were solved with an independently implemented CTMC at 161 and 321 grid points and with a separate Euler–Maruyama reflected-diffusion simulation.

The independent 161-point CTMC differs from the archived 161-point objective by median below  $10^{-4}\%$  and maximum 0.608%. The 161-to-321 grid change is median 1.17% and maximum 2.25%. The Monte Carlo/321-point CTMC difference is median 2.52%, 90th percentile 4.94%, and maximum 6.67%. Median policy-rank Spearman is 1.0, and the best policy agrees in 83.3% of scenario comparisons.

The campaign strengthens model and implementation validity but does not remove the absence of end-to-end controller execution on real infrastructure.

# 9 Reproducibility map

The repository archive contains:

- simulator and optimizer code;
- exhaustive 190-policy training and holdout surfaces;
- comparator and ablation outputs;

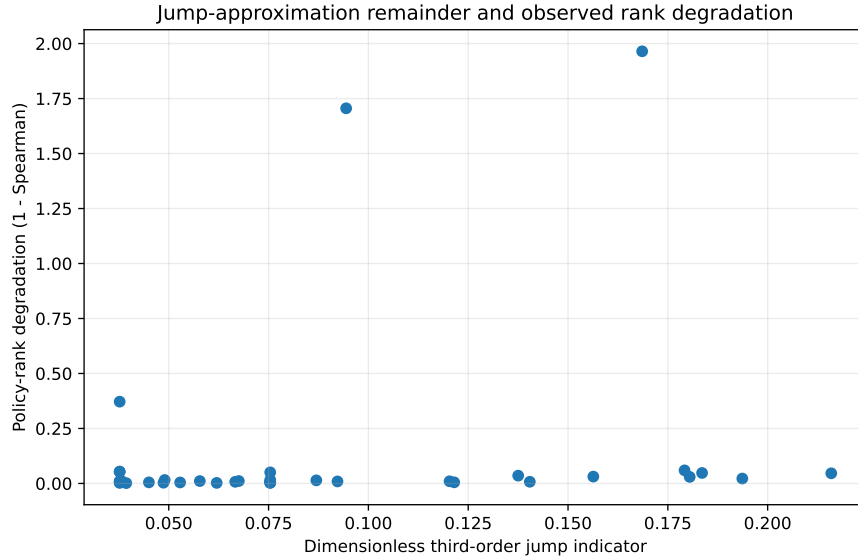


Figure 7: Third-order jump indicator and policy-rank degradation.

- trace transformation and replay scripts;
- original and seven-day protocol locks;
- fresh low-load confirmation outputs;
- source-trace provenance hashes;
- calibration data and scripts;
- figures, result tables, and checksum manifests; and
- audit scripts and human-readable reports.

The code, data, audit artifacts, and reproducibility materials are available at

[doi:10.5281/zenodo.21434924](https://doi.org/10.5281/zenodo.21434924).

The corresponding GitHub repository and release are

[AMD-Aljohani/dr-bgs-llm-routing](https://github.com/AMD-Aljohani/dr-bgs-llm-routing)

and

[v1.1.0](https://github.com/AMD-Aljohani/dr-bgs-llm-routing/releases/tag/v1.1.0).

Immutable provenance checks validate tracked lock metadata, locked-input hashes, required scientific artifacts, and the repository checksum manifest. The repository does not claim that historical filesystem modification times prove execution chronology.

## 10 Supplementary interpretation

The expanded material supports the same restrained conclusion as the main manuscript. DR-BGS is a problem-specific simulation-optimization workflow, not a general safety theorem. Structural search coverage contributes most of the finite-budget reliability, while the diffusion residual provides an additional ranking improvement. Independent later-pool risk evaluation can abstain even when nomination is near exhaustive selection. Exhaustive high-load evaluation and enlarged certification batches show that the reported abstention is not explained by a missed tested policy or the original sample size. Positive low-load confirmation identifies a modeled operating envelope rather than a recommended production configuration.

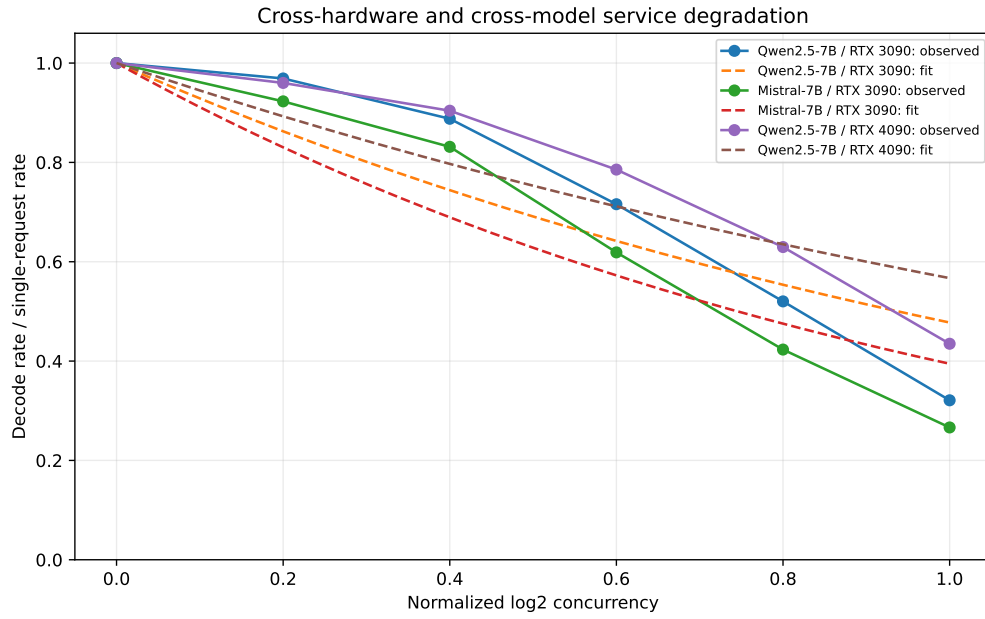


Figure 8: Normalized service curves and domain-specific exponential fits.

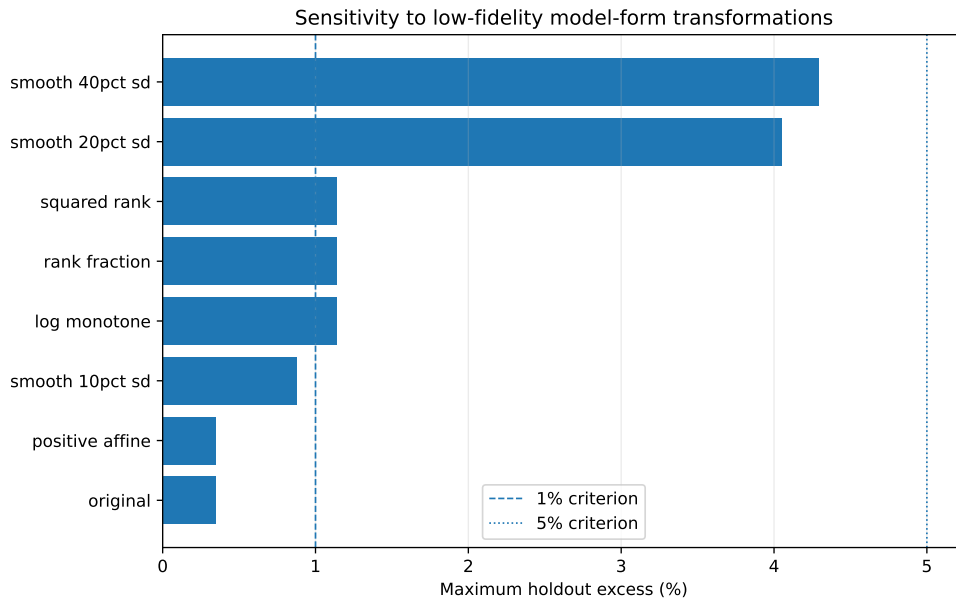


Figure 9: Maximum holdout excess after low-fidelity transformations.

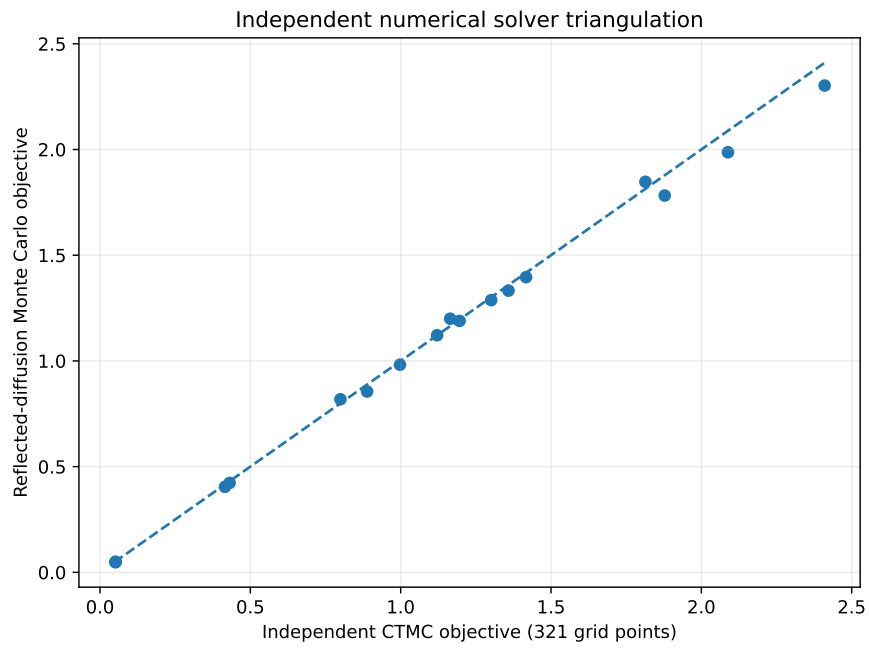


Figure 10: Independent CTMC and stochastic reflected-diffusion objectives.